Vraj Patel

Software Engineering | Backend, Full-Stack and Internal Tools

Boulder, Colorado | U.S. Citizen | vrajmpatel.com | linkedin.com/in/vrajmp
GitHub: basechildren | PatVraj

EXPERIENCE

Software Engineer, Internal Tools May 2025 - Present
CU Boulder | Institute of Behavioral Science

- Build a FastAPI, React/TypeScript, and PostgreSQL support-operations platform around 10,803 reconciled Freshdesk tickets in the August 2026 snapshot.
- Designed incremental source reconciliation with durable watermarks, idempotent upserts, lifecycle-aware records, and fail-closed readiness checks.
- Reduced the analytics visualization payload from 361 KB to 123 KB gzip (66%) by replacing Plotly with Recharts and accessible chart tables.
- Automated Active Directory offboarding with PowerShell, reducing typical group-cleanup effort from about 15 minutes to under one minute per user.

Research Assistant, Data Engineering Jan 2025 - May 2026
Princeton University | Indian parliamentary research

- Built fault-tolerant Selenium collection workflows for Indian parliamentary archives spanning 1951-2024, with pagination and resumable SQLite state.
- Extracted thousands of parliamentary PDFs with PyMuPDF and standardized inconsistent ministry names with fuzzy matching and reviewable intermediate data.

Lead Technical Research Assistant Mar 2023 - May 2025
CU Boulder | Institute of Behavioral Science

- Built a Python/AWS speech pipeline using Amazon Transcribe, Translate, and S3; the combined campaign collection covered 200+ hours, with 181 speeches in the 2024 run.

SELECTED PROJECTS

SeeMyRace | Race Photo Search
Flask, PostgreSQL/pgvector, Docker

- Built authenticated, user-scoped match verification and GPX-backed race creation for a team race-photo application; authored 32 commits across 24 paths through seven merged PRs.

NetPrice Finder | Price Comparison
FastAPI, Redis, Playwright, Docker Compose

- Built a multi-service price-comparison system with a REST API, browser scraper, Redis caching, database persistence, container health checks, and an internal service network.

EDUCATION

M.S. Computer Science + M.E. Engineering Management Expected May 2028
University of Colorado Boulder | Concurrent dual degree

B.S. Computer Science, Business Minor May 2025
University of Colorado Boulder | GPA: 3.84 technical, 3.65 overall

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C++, Bash, PowerShell
Software: FastAPI, Flask, React, SQLAlchemy, REST, gRPC, pytest
Data: PostgreSQL, SQLite, Redis, pgvector, Pandas, source reconciliation, data validation
Infrastructure: Docker, Kubernetes/OpenShift, AWS, GCP, Terraform, Linux, GitHub Actions